

Program: Diplomain Communication and Computer Networking	
Course Code: 3329	Course Title: Logic Design Lab
Semester: 3	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To understand digital logic and create various systems by using these logic

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Implement basic logic gates	8	Applying
CO2	Design and Implement Combinational Circuits	13	Applying
CO3	Demonstrate synchronous sequential logic	16	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	3			3		
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	Implement basic logic gates		
M1.01	Verify the Logic behavior of various IC gates: 7408, 7432, 7404, 7400, 7402, 7486)	4	Applying
M1.02	Implement basic gates using Universal Gates (NAND & NOR)	4	Applying
CO2	Design and Implement Combinational Circuits		
M2.01	Implement the following Combinational Circuits (Design, construct, and test combinational logic circuit that generates parity bit from four message bits)	5	Applying
M2.02	Design a combinational circuit that converts a Gray code to binary, Decoder for a binary digit to BCD, and a seven segment indicator	4	Applying
M2.03	Design, construct, and test a half-adder & a full-adder Implement using Basic gates and Universal gates	4	Applying
	SeriesTest-I	2	
CO3	Demonstrate synchronous sequential logic.		
M3.01	Construct, Test, and investigate the operation of SR Flip-Flop, JK Flip-Flop, D Flip-Flop, T Flip-Flop, Master-Slave JK Flip-Flop.	6	Applying
M3.02	Sequential Circuits (Design, construct, and test Synchronous Up-Down counter with Enable)	4	Applying
M3.03	Construct and test Ripple Counter, Synchronous counter	6	Applying
	SeriesTest-II	2	
	Openendedexperiment/microprojects	4	

Note: Experiments shall be conducted such that all COs are attained.
 Minimum of 6 experiments (excluding open ended experiments/micro projects) shall be performed.
 Compulsory for CIA
 The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text/Reference

T/R	BookTitle/Author
T1	Thomas L Floyd, Digital Fundamentals, 10/e, Pearson Education, 2009
R1	M. Morris Mano, Digital Logic & Computer Design, 4/e, Pearson Education, 2013
R2	Donald D Givone, Digital Principles and Design, Tata McGraw Hill, 2003
R3	Thomas L Floyd, Digital Fundamentals, 10/e, Pearson Education, 2009

Online Resources

Sl.No	WebsiteLink
1	http://nptel.ac.in/course:s/Webcourse-contents/IIT-%20Guwahati/comp_org_arc/web/
2	https://www.tutorialspoint.com/computer_logical_organization/combinational_circuits.htm
3	https://www.electronics-tutorials.ws/sequential/seq_6.html
4	https://www.tutorialspoint.com/digital_circuits/digital_circuits_boolean_algebra.htm